

Forest Inventory Report

North Fork Deer Creek Timber Harvesting Plan - Section 31, T17N R10E MDB&M; Nevada County, California

Prepared by William Steinley | September 06, 2026 | BAF 20 cruise, 142 plots on a 400 ft grid, 534 acres in 7 harvest units

DEMONSTRATION. The cruise data in this report are simulated; no trees were measured. The plan area, harvest units and terrain are real public data (see the companion THP map packet); the inventory workflow, statistics and report structure are the deliverable.

Summary

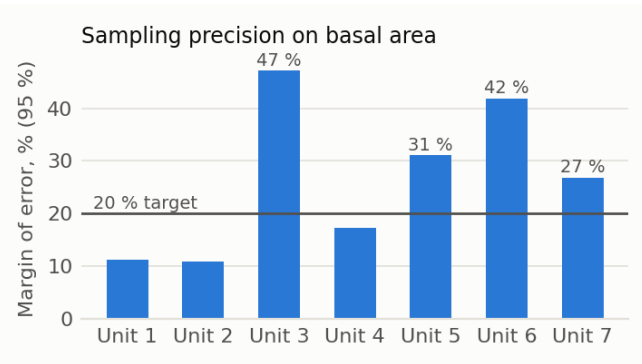
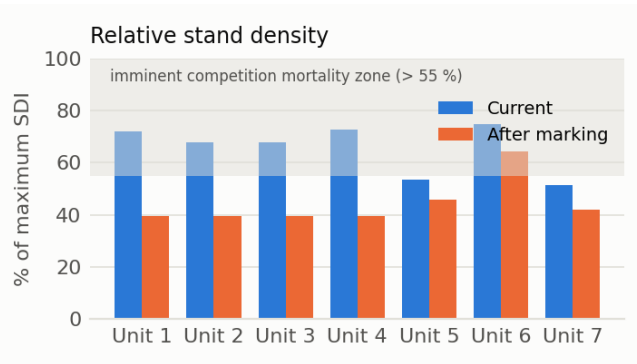
Acres cruised	Plots	Basal area sq ft/ac	QMD in	Trees / ac	% of max SDI	Net MBF standing	Harvest MBF net
534	142	234	12.2	288	67 %	15,227	2,078

The property carries 234 sq ft of basal area per acre at a quadratic mean diameter of 12.2 in, which puts it at 67 % of maximum stand density index, well inside the zone where competition mortality is expected. The four commercial-thin units (1-4) are the densest at 68-73 % of maximum; the cable-yarded selection units on the steeper ground (5-7) run 51-75 %. Standing net sawtimber is 15,227 MBF Scribner; a simulated marking (thin from below to 40 % of maximum SDI in the thinning units, species-weighted single-tree selection in the selection units) removes 2,078 MBF net (14 % of standing volume). Ponderosa pine and white fir dominate the basal area; sugar pine and black oak are retained under both prescriptions.

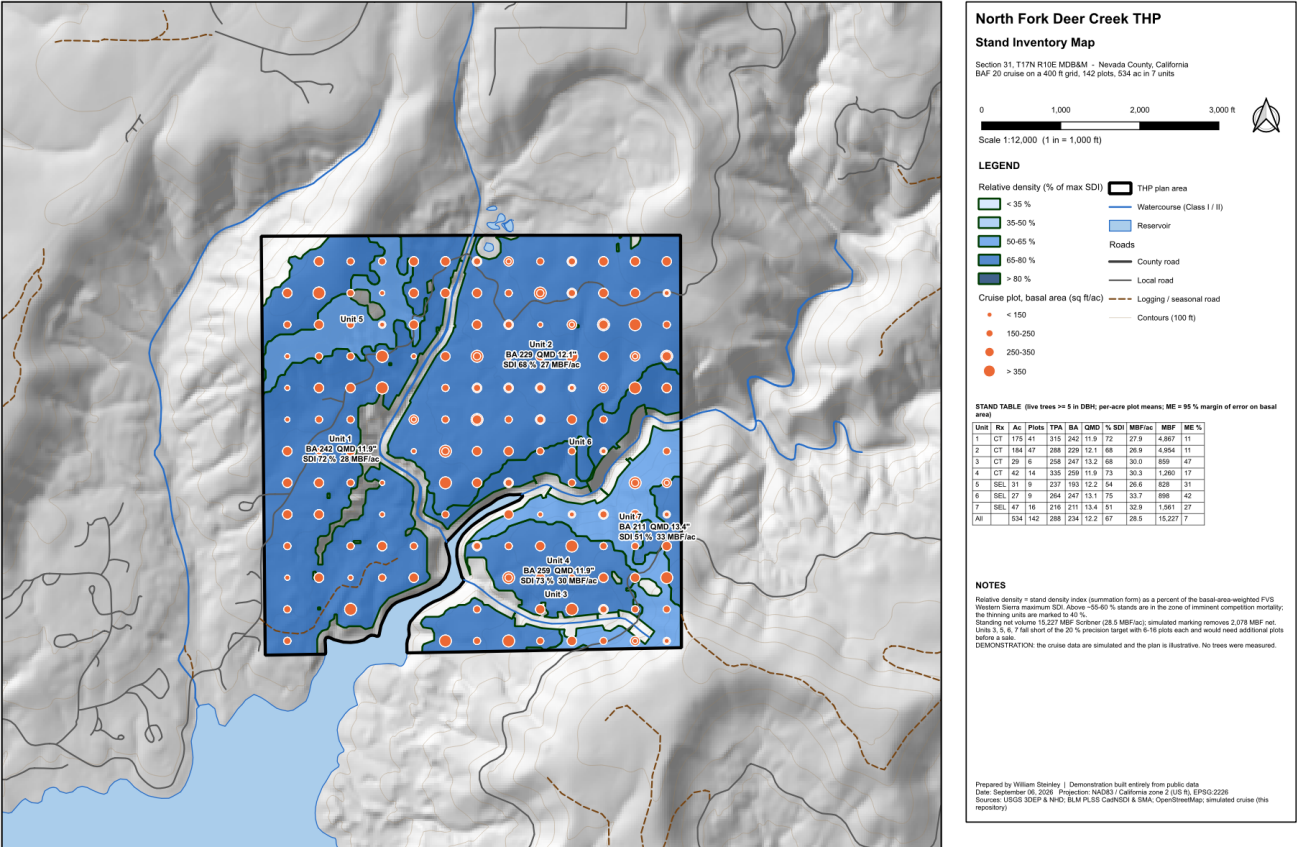
Sampling precision on basal area is 6.5 % at 95 % confidence for the property as a whole, but units 3, 5, 6, 7 exceed the 20 % target with only 6-16 plots each; another 85 plots across those units would bring them within target before a timber sale.

Unit	Rx	Acres	Plots	TPA	BA	QMD	SDI	% max	Net MBF/ac	Net MBF	Cut MBF	ME %
1	Thin	175	41	315	242	11.9	379	72	27.9	4,867	606	11
2	Thin	184	47	288	229	12.1	357	68	26.9	4,954	420	11
3	Thin	29	6	258	247	13.2	377	68	30.0	859	129	47
4	Thin	42	14	335	259	11.9	405	73	30.3	1,260	172	17
5	Select	31	9	237	193	12.2	289	54	26.6	828	170	31
6	Select	27	9	264	247	13.1	366	75	33.7	898	187	42
7	Select	47	16	216	211	13.4	306	51	32.9	1,561	394	27
All		534	142	288	234	12.2	361	67	28.5	15,227	2,078	7

Live trees ≥ 5 in DBH; per-acre values are plot means; ME = 95 % margin of error on basal area per acre as a percent of the mean.

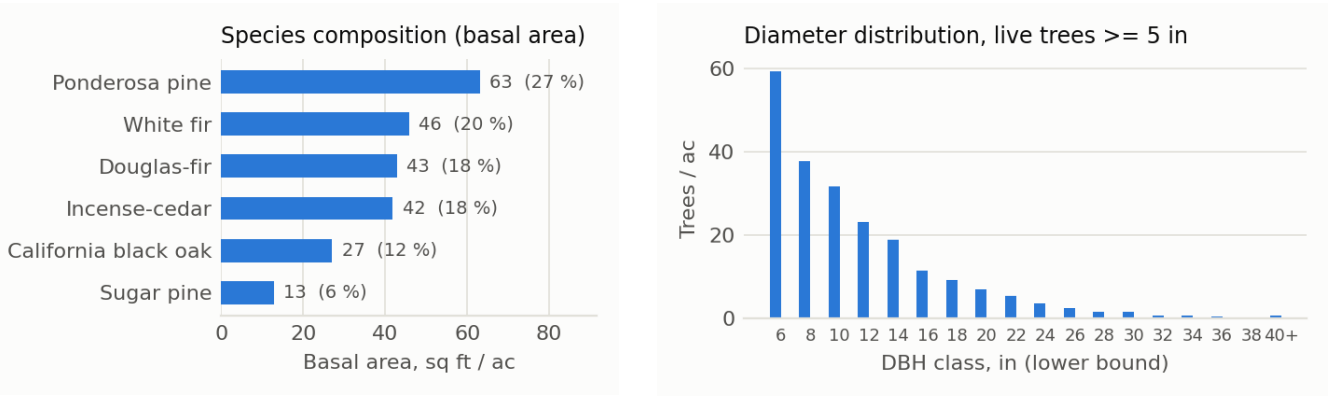


Stand inventory map



Units shaded by percent of maximum SDI; plots sized by basal area per acre. Full-size 11 x 17 sheet: Stand_Inventory_Map.pdf.

Property-wide structure



Methods

Design. Systematic 400 ft grid with a random start, 142 plots over 534 ac (one plot per 3.8 ac). Variable-radius prism plot, BAF 20, for trees ≥ 5 in DBH; 1/100-acre fixed-radius subplot for 1.0-4.9 in. Tree expansion factor = $BAF / (0.005454 \times DBH^2)$ on the prism plot and 100 on the subplot.

Stand metrics. Per-acre values are computed for every plot and averaged within each unit; property values are acre-weighted unit means. QMD = $\sqrt{BA / TPA / 0.005454}$. SDI uses the summation form, sum of $TPA \times (DBH/10)^{1.6}$, which is appropriate for the uneven diameter distributions here. Maximum SDI is the basal-area-weighted species maximum from the FVS Western Sierra Nevada variant (white fir 710, Douglas-fir 590, sugar pine 600, ponderosa pine / incense-cedar / black oak 450).

Volume. Merchantable cubic volume = form factor (0.42 conifers, 0.38 oak) $\times$ tree basal area $\times$ total height $\times$ 0.90, for live trees ≥ 10 in DBH. Scribner board feet = cubic feet $\times$ a ratio rising from 3.8 at 10 in to 6.2 at 36 in and above; net volume deducts the recorded defect percent. These are transparent approximations chosen for a demonstration; a production cruise would substitute regional volume or taper equations (for example NVEL Region 5 profiles).

Precision. Coefficient of variation and 95 % margin of error (t-distribution, n - 1 df) are computed on per-plot basal area and net volume. Plots needed for a given allowable error E are $(t \times CV / E)^2$. The 20 % target is the usual standard for a THP-level cruise.

Quality control. Duplicate records are dropped; DBH outside 1-80 in is excluded; heights over 250 ft or outside 0.5-1.6 $\times$ the species height-diameter curve (fitted to the clean data) are replaced by the curve estimate; missing or unknown species codes are carried as UNK in density metrics and excluded from species tables. All flags are listed in the appendix.

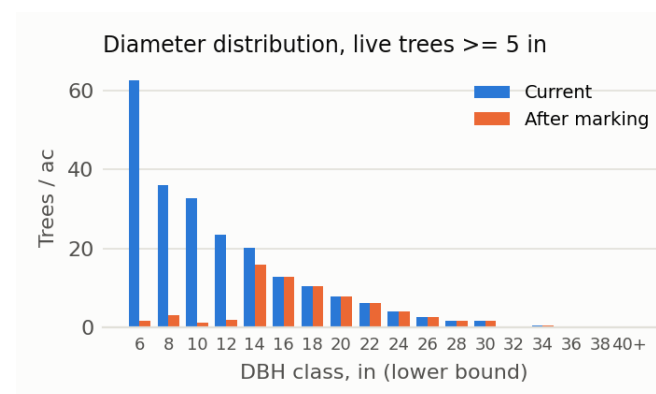
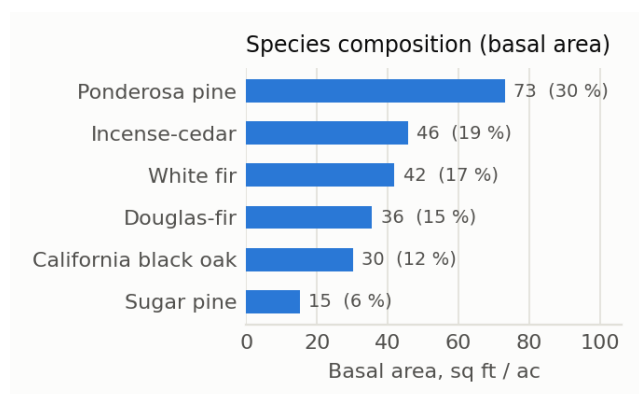
Prescriptions. Commercial-thin units are thinned from below to 40 % of maximum SDI, retaining all sugar pine and all black oak ≥ 16 in. Selection units are marked by species-weighted random selection of trees ≥ 12 in (about 40 % of white fir and incense-cedar basal area, 18 % of pine and Douglas-fir, none of the sugar pine or black oak). The marking is a simulation to show pre- and post-harvest structure and a harvest volume estimate; it is not an RPF's prescription.

Sampling statistics

Unit	Plots	Variable	Mean	SD	CV %	ME %	95 % CI	Plots for 10 %	Plots for 20 %
1	41	BA sq ft/ac	242	87	36	11.3	215 - 270	53	14
1	41	Net bf/ac	27,868	9,884	35	11.2	24,749 - 30,988	52	13
2	47	BA sq ft/ac	229	86	37	11.0	204 - 255	57	15
2	47	Net bf/ac	26,863	10,268	38	11.2	23,848 - 29,878	60	15
3	6	BA sq ft/ac	247	111	45	47.1	130 - 363	134	34
3	6	Net bf/ac	29,975	14,684	49	51.4	14,565 - 45,386	159	40
4	14	BA sq ft/ac	259	78	30	17.4	214 - 304	43	11
4	14	Net bf/ac	30,330	12,249	40	23.3	23,258 - 37,403	77	20
5	9	BA sq ft/ac	193	78	40	31.1	133 - 253	87	22
5	9	Net bf/ac	26,589	10,856	41	31.4	18,245 - 34,933	89	23
6	9	BA sq ft/ac	247	134	54	41.8	144 - 350	158	40
6	9	Net bf/ac	33,727	17,595	52	40.1	20,202 - 47,253	145	37
7	16	BA sq ft/ac	211	107	50	26.9	154 - 268	116	29
7	16	Net bf/ac	32,892	17,875	54	29.0	23,367 - 42,417	135	34
All	142	BA sq ft/ac	234	92	39	6.5	218 - 249	61	16
All	142	Net bf/ac	28,724	12,140	42	7.0	26,710 - 30,738	70	18

Unit 1 - Commercial Thin, tractor yarding, 175 acres

Plots	Trees / ac	Basal area	QMD	SDI	% max SDI	Net MBF/ac	Net MBF	Snags / ac	Regen / ac
41	315	242	11.9 in	379	72 %	27.9	4,867	16.2	217



Size-class structure (live trees, per acre)

Size class	Trees / ac	BA sq ft/ac	Net bf/ac
5-11.9 in (poles)	223	67	1,424
12-23.9 in (small sawtimber)	81	123	14,791
24-35.9 in (large sawtimber)	11	44	9,344
36 in + (very large)	1	8	2,309

Simulated marking

Thin from below to 40 % of max SDI; retain all sugar pine and black oak >= 16 in.

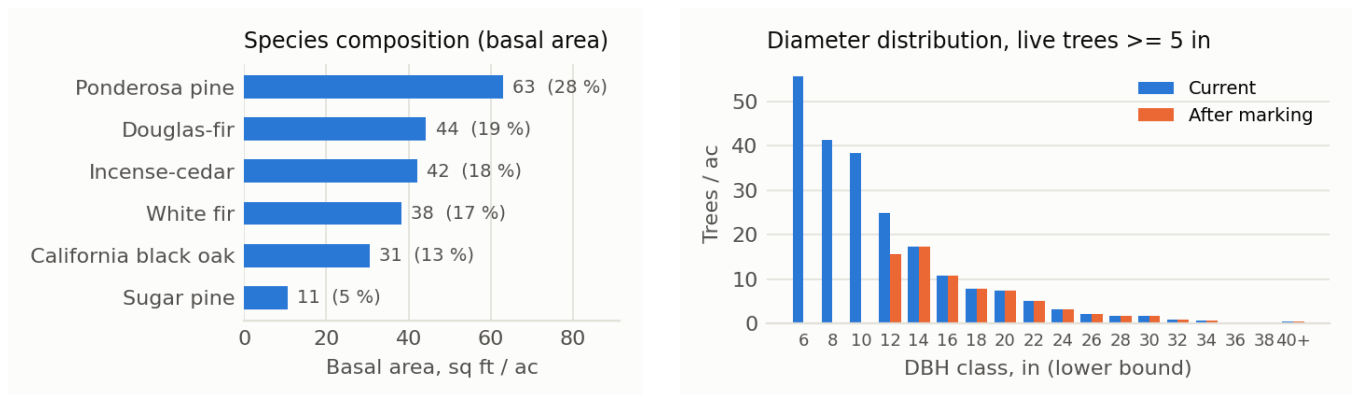
	Trees / ac	BA sq ft/ac	QMD in	SDI	% max SDI	Net MBF/ac
Before	315	242	11.9	379	72	27.9
Removed	243	88 (36 %)	8.2			3.5
After	73	154	19.7	210	40	24.4

Harvest: 606 MBF net (650 gross) on 175 ac; by species: Ponderosa pine 202, Douglas-fir 120, White fir 114, Incense-cedar 88, California black oak 73, Unknown 8.

Precision: basal area CV 36 %, margin of error 11.3 % with 41 plots - meets the target.

Unit 2 - Commercial Thin, tractor yarding, 184 acres

Plots	Trees / ac	Basal area	QMD	SDI	% max SDI	Net MBF/ac	Net MBF	Snags / ac	Regen / ac
47	288	229	12.1 in	357	68 %	26.9	4,954	13.3	209



Size-class structure (live trees, per acre)

Size class	Trees / ac	BA sq ft/ac	Net bf/ac
5-11.9 in (poles)	203	68	1,715
12-23.9 in (small sawtimber)	73	107	12,797
24-35.9 in (large sawtimber)	10	46	9,623
36 in + (very large)	1	9	2,729

Simulated marking

Thin from below to 40 % of max SDI; retain all sugar pine and black oak >= 16 in.

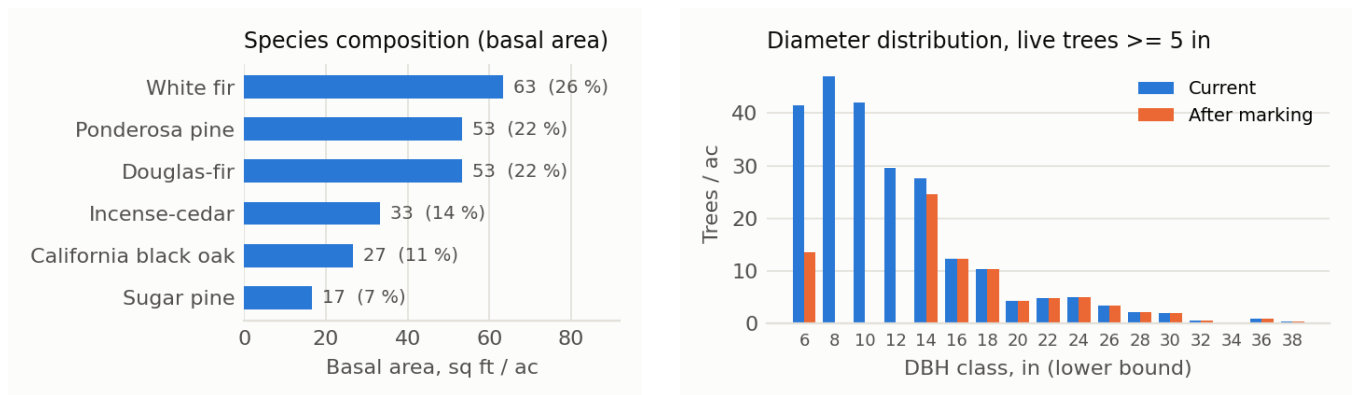
	Trees / ac	BA sq ft/ac	QMD in	SDI	% max SDI	Net MBF/ac
Before	288	229	12.1	357	68	26.9
Removed	212	75 (33 %)	8.1			2.3
After	75	154	19.4	210	39	24.6

Harvest: 420 MBF net (455 gross) on 184 ac; by species: Ponderosa pine 127, Incense-cedar 85, White fir 74, Douglas-fir 69, California black oak 66.

Precision: basal area CV 37 %, margin of error 11.0 % with 47 plots - meets the target.

Unit 3 - Commercial Thin, tractor yarding, 29 acres

Plots	Trees / ac	Basal area	QMD	SDI	% max SDI	Net MBF/ac	Net MBF	Snags / ac	Regen / ac
6	258	247	13.2 in	377	68 %	30.0	859	12.3	183



Size-class structure (live trees, per acre)

Size class	Trees / ac	BA sq ft/ac	Net bf/ac
5-11.9 in (poles)	155	60	1,859
12-23.9 in (small sawtimber)	89	123	14,098
24-35.9 in (large sawtimber)	13	53	11,156
36 in + (very large)	1	10	2,863

Simulated marking

Thin from below to 40 % of max SDI; retain all sugar pine and black oak >= 16 in.

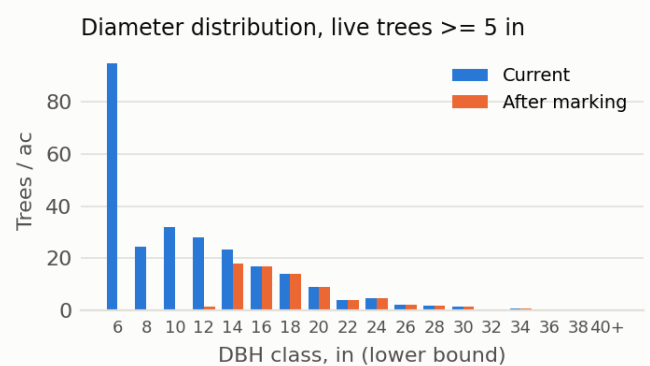
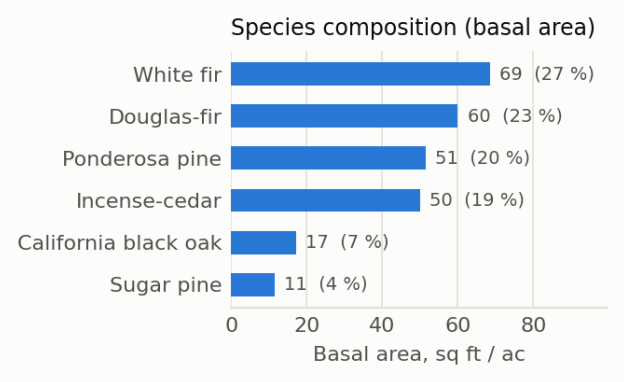
	Trees / ac	BA sq ft/ac	QMD in	SDI	% max SDI	Net MBF/ac
Before	258	247	13.2	377	68	30.0
Removed	174	87 (35 %)	9.6			4.5
After	84	160	18.7	219	40	25.5

Harvest: 129 MBF net (140 gross) on 29 ac; by species: White fir 45, Ponderosa pine 36, Douglas-fir 28, California black oak 14, Incense-cedar 6.

Precision: basal area CV 45 %, margin of error 47.1 % with 6 plots - 28 more plots needed to reach the 20 % target.

Unit 4 - Commercial Thin, tractor yarding, 42 acres

Plots	Trees / ac	Basal area	QMD	SDI	% max SDI	Net MBF/ac	Net MBF	Snags / ac	Regen / ac
14	335	259	11.9 in	405	73 %	30.3	1,260	8.9	229



Size-class structure (live trees, per acre)

Size class	Trees / ac	BA sq ft/ac	Net bf/ac
5-11.9 in (poles)	228	66	1,539
12-23.9 in (small sawtimber)	95	140	17,038
24-35.9 in (large sawtimber)	11	47	9,966
36 in + (very large)	1	6	1,788

Simulated marking

Thin from below to 40 % of max SDI; retain all sugar pine and black oak ≥ 16 in.

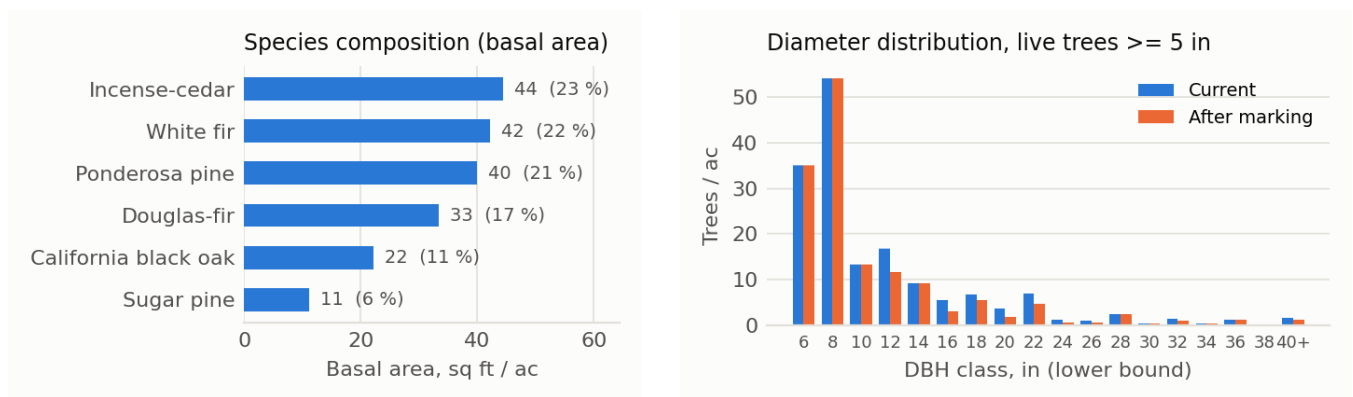
	Trees / ac	BA sq ft/ac	QMD in	SDI	% max SDI	Net MBF/ac
Before	335		11.9	405	73	30.3
Removed	260	96 (37 %)	8.2			4.1
After	75	163	19.9	222	40	26.2

Harvest: 172 MBF net (185 gross) on 42 ac; by species: White fir 60, Ponderosa pine 48, Incense-cedar 26, Douglas-fir 22, California black oak 16.

Precision: basal area CV 30 %, margin of error 17.4 % with 14 plots - meets the target.

Unit 5 - Selection, cable yarding, 31 acres

Plots	Trees / ac	Basal area	QMD	SDI	% max SDI	Net MBF/ac	Net MBF	Snags / ac	Regen / ac
9	237	193	12.2 in	289	54 %	26.6	828	31.4	100



Size-class structure (live trees, per acre)

Size class	Trees / ac	BA sq ft/ac	Net bft/ac
5-11.9 in (poles)	177	51	675
12-23.9 in (small sawtimber)	49	78	9,372
24-35.9 in (large sawtimber)	7	33	7,369
36 in + (very large)	3	31	9,174

Simulated marking

Single-tree selection, trees >= 12 in: remove ~40 % of white fir / incense-cedar BA, ~18 % of pine / Douglas-fir; retain sugar pine and black oak.

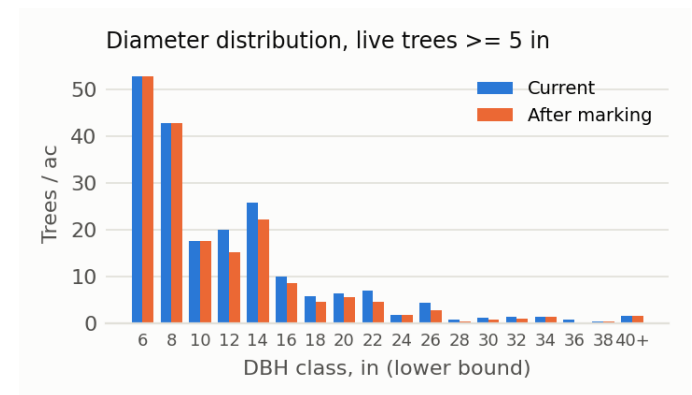
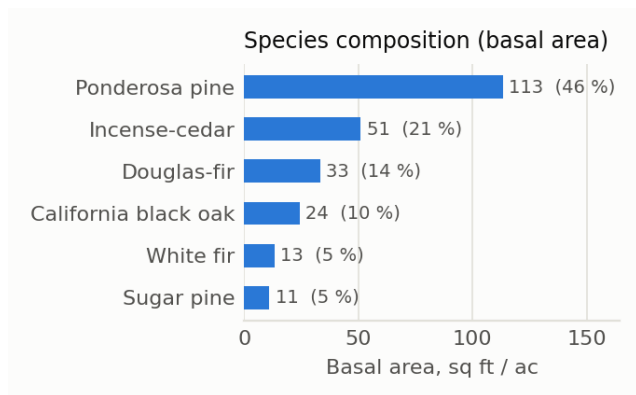
	Trees / ac	BA sq ft/ac	QMD in	SDI	% max SDI	Net MBF/ac
Before	237		12.2	289	54	26.6
Removed	15	33 (17 %)	20.4			5.5
After	222	160	11.5	245	46	21.1

Harvest: 170 MBF net (183 gross) on 31 ac; by species: White fir 74, Incense-cedar 60, Douglas-fir 18, Ponderosa pine 18.

Precision: basal area CV 40 %, margin of error 31.1 % with 9 plots - 13 more plots needed to reach the 20 % target.

Unit 6 - Selection, cable yarding, 27 acres

Plots	Trees / ac	Basal area	QMD	SDI	% max SDI	Net MBF/ac	Net MBF	Snags / ac	Regen / ac
9	264	247	13.1 in	366	75 %	33.7	898	15.1	78



Size-class structure (live trees, per acre)

Size class	Trees / ac	BA sq ft/ac	Net bf/ac
5-11.9 in (poles)	174	53	820
12-23.9 in (small sawtimber)	75	111	13,271
24-35.9 in (large sawtimber)	11	53	11,249
36 in + (very large)	3	29	8,387

Simulated marking

Single-tree selection, trees >= 12 in: remove ~40 % of white fir / incense-cedar BA, ~18 % of pine / Douglas-fir; retain sugar pine and black oak.

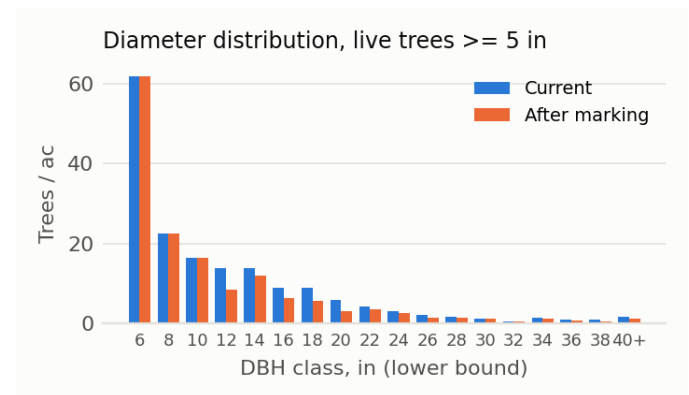
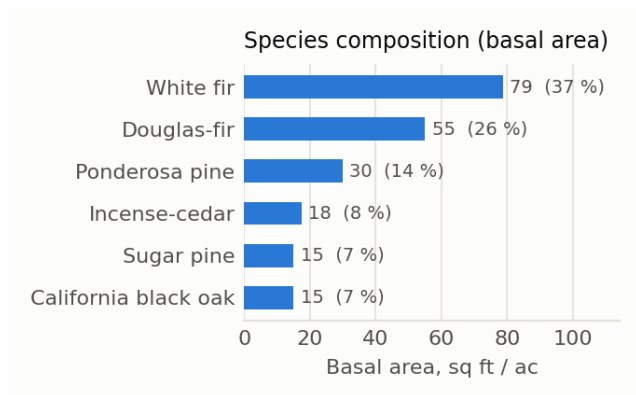
	Trees / ac	BA sq ft/ac	QMD in	SDI	% max SDI	Net MBF/ac
Before	264	247	13.1	366	75	33.7
Removed	18	40 (16 %)	20.3			7.0
After	246	207	12.4	312	64	26.7

Harvest: 187 MBF net (198 gross) on 27 ac; by species: Ponderosa pine 104, Incense-cedar 37, White fir 28, Douglas-fir 17.

Precision: basal area CV 54 %, margin of error 41.8 % with 9 plots - 31 more plots needed to reach the 20 % target.

Unit 7 - Selection, cable yarding, 47 acres

Plots	Trees / ac	Basal area	QMD	SDI	% max SDI	Net MBF/ac	Net MBF	Snags / ac	Regen / ac
16	216	211	13.4 in	306	51 %	32.9	1,561	10.0	75



Size-class structure (live trees, per acre)

Size class	Trees / ac	BA sq ft/ac	Net bf/ac
5-11.9 in (poles)	147	42	856
12-23.9 in (small sawtimber)	56	88	10,842
24-35.9 in (large sawtimber)	10	46	10,568
36 in + (very large)	4	35	10,626

Simulated marking

Single-tree selection, trees >= 12 in: remove ~40 % of white fir / incense-cedar BA, ~18 % of pine / Douglas-fir; retain sugar pine and black oak.

	Trees / ac	BA sq ft/ac	QMD in	SDI	% max SDI	Net MBF/ac
Before	216	211	13.4	306	51	32.9
Removed	20	45 (21 %)	20.6			8.3
After	197	166	12.4	246	42	24.6

Harvest: 394 MBF net (418 gross) on 47 ac; by species: White fir 252, Douglas-fir 73, Incense-cedar 45, Ponderosa pine 24.

Precision: basal area CV 50 %, margin of error 26.9 % with 16 plots - 13 more plots needed to reach the 20 % target.

Appendix A - Data quality flags

2019 tree records checked: 1 duplicate dropped, 1 record excluded, 2 heights replaced from the height-diameter curve.

Row	Plot	Severity	Field	Value	Issue	Action
260	U1-19	CRITICAL	dbh	320.0	DBH 320.0 in outside 1-80 in	Tree excluded from all metrics
484	U1-34	CRITICAL	height_ft	265.0	Height 265 ft exceeds 250 ft	Height replaced by H-D curve estimate
553	U1-38	CRITICAL	species	XX	Species code 'XX' not in code list	Set to UNK; kept in density metrics, excluded from species tables
817	U2-14	CRITICAL	species		Species missing	Set to UNK; kept in density metrics, excluded from species tables
1979	U7-13	CRITICAL	(row)		Duplicate tree record	Dropped duplicate
1776	U6-07	WARNING	height_ft	18.0	Height 18 ft vs 75 ft expected for a 25.2 in BO	Height replaced by H-D curve estimate

Appendix B - Height-diameter curves fitted to the cruise

Species	a	b	Trees
California black oak	1.473	0.864	243
Douglas-fir	1.800	0.880	369
Incense-cedar	1.191	1.014	359
Ponderosa pine	1.675	0.904	537
Sugar pine	1.781	0.880	111
White fir	1.665	0.919	395

$\ln(H - 4.5) = a + b \ln(DBH)$, H in feet, DBH in inches.

Data sources and software

Harvest units, plan area, roads and watercourses from the North Fork Deer Creek THP map packet (USGS 3DEP and NHD, BLM PLSS and SMA, OpenStreetMap). Cruise data simulated by scripts/02_simulate_cruise.py in this repository. Analysis in Python 3.12 (pandas, numpy, scipy, geopandas, openpyxl, matplotlib, ReportLab) run through QGIS 3.44; map in PyQGIS. Source and outputs: github.com/woodsylwill/forest-inventory-dashboard